

HYDABURG, AK, USA

WMO: 703884

Lat: 55.206N

Lon: 132.828W

Elev: 0

StdP: 101.33

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25376

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-5.2	-17.0	0.8	-4.2	-13.0	1.2	-3.1	17.6	7.8	15.5	7.2	1.8	20	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	21.3	14.9	19.1	13.6	17.7	13.0	15.6	19.5	14.7	17.9	14.0	16.8	4.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.7	9.8	16.8	12.9	9.3	15.9	12.5	9.0	15.4	43.3	19.9	40.8	17.8	39.1	16.7	18.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	11.7	9.7	-10.5	22.8	2.7	3.1	-12.5	25.0	-14.1	26.8	-15.6	28.5	-17.5	30.7
			-10.6	16.3	3.0	1.2	-12.7	17.1	-14.5	17.9	-16.2	18.5	-18.4	19.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	2.9	3.3	3.8	6.0	8.7	11.8	13.8	14.2	11.8	8.8	5.1	3.4
	DBStd	4.88	4.05	2.90	2.42	2.11	2.26	2.07	1.79	1.62	2.35	2.96	2.91	3.39
	HDD10.0	1194	220	188	193	122	53	3	0	0	8	55	148	204
	HDD18.3	3840	478	421	451	369	298	197	142	130	195	297	398	463
	CDD10.0	397	0	0	0	2	13	57	117	129	62	17	1	0
	CDD18.3	3	0	0	0	0	0	1	1	1	0	1	0	0
	CDH23.3	17	0	0	0	0	0	8	7	3	0	0	0	0
	CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0
Wind	WSAvg	3.4	3.7	3.4	3.6	3.4	3.1	3.0	2.8	2.7	3.3	3.7	3.7	3.9
Precipitation	PrecAvg	2661	293	218	213	163	117	96	121	171	271	341	328	329
	PrecMax	3625	544	451	401	249	293	179	225	372	489	531	587	716
	PrecMin	1912	89	78	76	16	8	35	11	43	100	149	151	151
	PrecStd	433	91	100	84	58	66	36	52	77	93	86	105	137
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	12.0	12.0	17.7	19.8	23.8	24.0	23.0	19.7	20.9	12.6	11.1
		MCWB	5.9	5.9	6.9	9.0	11.5	14.6	16.9	16.0	13.5	13.9	7.8	7.7
	2%	DB	9.1	9.0	9.0	13.0	16.7	19.3	21.1	21.0	17.5	18.7	11.0	9.1
		MCWB	6.8	6.3	5.9	7.2	10.2	13.2	15.4	15.3	12.7	11.8	8.3	7.1
	5%	DB	8.2	7.9	7.9	11.1	14.1	17.4	18.6	18.7	16.2	13.0	9.7	8.7
		MCWB	6.6	5.7	5.4	6.8	9.3	12.3	14.1	14.4	12.4	10.5	7.8	7.0
	10%	DB	7.7	7.4	7.5	9.1	12.6	15.2	17.2	17.3	14.9	12.3	8.8	7.7
		MCWB	6.1	5.2	5.2	6.1	8.9	11.4	13.2	13.5	12.1	10.1	7.3	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	8.0	7.9	10.4	12.6	15.2	17.3	16.8	15.1	13.6	9.8	8.5
		MCDB	8.6	9.3	9.6	14.7	17.1	20.7	23.3	21.8	17.5	19.0	10.8	9.3
	2%	WB	7.8	7.1	7.1	8.4	10.9	13.7	15.7	15.6	13.9	12.1	8.6	7.9
		MCDB	8.3	8.2	8.4	11.9	14.7	18.7	20.1	19.3	16.0	15.4	9.6	8.6
	5%	WB	7.2	6.1	6.2	7.6	10.2	12.7	14.5	14.9	13.2	11.3	7.9	7.3
		MCDB	8.0	7.9	7.9	9.8	13.4	16.3	17.7	17.8	15.3	13.9	8.9	8.0
	10%	WB	6.3	5.3	5.3	6.8	9.4	11.8	13.7	14.2	12.6	10.4	7.3	6.3
		MCDB	7.7	7.4	7.3	8.9	12.3	14.6	16.4	16.8	14.5	12.2	8.5	7.7
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.6	4.9	5.8	5.7	5.3	4.9	5.1	5.1	4.4	4.1	3.9
		MCDBR	4.1	4.9	5.9	9.8	9.5	9.8	9.5	9.2	7.5	4.9	4.3	3.6
		MCWBR	3.3	3.5	4.0	5.3	5.0	4.9	4.9	4.8	4.3	3.6	3.5	2.8
	5% WB	MCDBR	3.7	4.0	4.7	7.0	7.7	8.1	7.7	7.1	5.5	3.7	4.1	3.5
		MCWBR	3.4	3.2	3.7	4.4	4.3	4.2	4.2	4.1	4.0	3.4	3.8	3.4
Clear-Sky Solar Irradiance	taub	0.306	0.316	0.341	0.360	0.366	0.364	0.374	0.361	0.361	0.361	0.338	0.313	0.307
	taud	2.361	2.475	2.420	2.394	2.426	2.470	2.475	2.536	2.489	2.462	2.329	2.242	
	Ebn at Noon	630	777	833	864	878	883	865	857	803	733	602	514	
	Edh at Noon	53	67	88	103	106	103	100	89	82	67	53	47	
All-Sky Solar Radiation	RadAvg	0.47	1.24	2.26	3.70	4.63	4.86	4.32	3.77	2.50	1.36	0.63	0.37	
	RadStd	0.06	0.22	0.31	0.23	0.45	0.42	0.52	0.38	0.32	0.17	0.08	0.06	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page